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Self-assembled monolayer-modified hole transport layers for high-performance CMOS-compatible PbS quantum dot photodetectors

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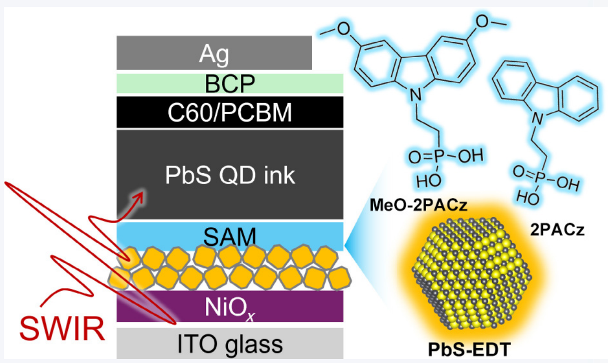
ABSTRACT: PbS colloidal quantum dots (QDs) show great promise for short-wave infrared (SWIR) photodetection due to their tunable photoresponse and cost-effective solution processability, positioning them as a strong competitor to InGaAs technologies. Inverted device architectures, essential for compatibility with complementary metal-oxide-semiconductor (CMOS) readout circuits, face performance challenges due to limitations in the hole transport layer (HTL), such as porous NiO_x structures that cause surface recombination at low annealing temperatures. To overcome these challenges, herein, we develop a multi-HTL strategy integrating NiO_x, 1,2-ethanedithiol (EDT)-treated PbS, and self-assembled monolayers (SAMs) including [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz) and [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz), significantly boosting the performance of inverted PbS QD photodetectors, with full-fullerene-based electron transport layers (ETLs). We confirm that the SAMs can effectively block electron transfer and passivate surface defects between the HTL and active layer, with 2PACz achieving an external quantum efficiency of 53% at 1200 nm and MeO-2PACz reducing dark current to 220 nA/cm^2, yielding a specific detectivity of 1.64×10^{12} Jones, which represents the highest reported value under similar testing conditions and in this spectral region. This multi-HTL strategy enables high-performance SWIR imaging compatible with CMOS and thin-film transistor (TFT) circuits, advancing QD-based photodetection technologies.



KEYWORDS: quantum dots, photodetector, complementary metal-oxide-semiconductor (CMOS) compatible, grazing-incidence small-angle X-ray scattering (GISAXS), interface

1 Introduction

Colloidal quantum dots (QDs) are promising materials for short-wave infrared (SWIR) range photodetection and imaging due to their tunable spectral response and solution processability, which enable low-cost fabrication and strong competitiveness with conventional InGaAs semiconductive techniques [1–7]. Among different QDs in recent literature, PbS QDs have attracted the most attention due to their low fabrication complexity and excellent device performance. In practical imaging applications, QDs are typically integrated into an inverted device architecture to ensure compatibility with the existing complementary metal-oxide-semiconductor (CMOS) based readout integrated circuits (ROICs), where the metal pads function as the bottom electrodes of the photodetectors. In this configuration, the hole transport layer (HTL) is deposited first, followed by the QD active layer and then

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the electron transport layer (ETL). Moreover, the inverted architecture facilitates more efficient charge carrier collection under a top-illumination configuration [8–11]. For the HTL, NiO_x nanoparticles are most commonly employed as the HTL [12]. However, at the low annealing temperatures typically used during fabrication, they tend to form porous structures, which can lead to substantial surface recombination. Notably, during imager chip fabrication, elevated temperatures are avoided to protect the underlying integrated circuits. As a result, inverted devices generally exhibit lower performance compared to their conventional counterparts, as widely reported in the literature.

Achieving high-quality QD-based SWIR imaging requires improving the performance of inverted devices, with a focus on enhancing external quantum efficiency (EQE), reducing dark current density (J_D), and minimizing noise currents. A typical inverted device structure consists of, from bottom to top: a bottom electrode, HTL, active layer, ETL, and a top electrode. The ETL via sputtering or atomic layer deposition (ALD) have been shown to significantly enhance the performance of inverted devices [11]. Nonetheless, the HTL still presents significant challenges. Besides the NiO_x HTL, thiol (e.g. 1,2-ethanedithiol (EDT)) treated QDs are frequently used as an additional HTL between the active layer and NiO_x HTL to fix the energy level alignment [8, 13]. Notably, the thiol-treated QD layer, upon partial oxidation, can exhibit p-type semiconductor properties and serve as an effective HTL in QD devices by forming a graded energy band structure that promotes hole extraction and suppresses undesired recombination [14–18]. To avoid the inhomogeneous morphology of the QD HTL caused by the solid-state ligand exchange, Wang et al. employed a monolayer of oxidized ZeTe as the HTL, resulting in improved performance of inverted PbS QD photodetectors [1].

Self-assembled monolayers (SAMs) are emerging HTL options in the perovskite field. They have been demonstrated to act as effective electron-blocking layers and efficiently passivate surface defects [19–23]. R.D. Campos et al. explored the use of SAM as a single HTL in inverted PbS QD solar cells and demonstrated their effectiveness in this role [24]. In field of inverted QD photodetectors, Ye et al. successfully employed a SAM to passivate the surface indium-tin oxide (ITO) bottom electrode. The SAM acted synergically with the Ag:HgTe QD layer to function as the HTL in Se:HgTe QD-based SWIR photodetectors [25]. All these efforts indicate SAMs offer a low-cost, solution-processable alternative to vacuum-based interfacial layers, with tunable molecular properties that enable precise energy-level alignment. These features make them highly suitable for large-scale integration in inverted QD photodetectors (circuit compatible) in a mild fabrication condition.

In this work, we investigated the effectiveness of two types of SAMs respectively as additional HTL in inverted PbS QD-based SWIR photodetectors. Our results demonstrate that incorporating a multilayer HTL structure (NiO_x/PbS-EDT/SAM) significantly enhances the performance of inverted QD photodetectors in a fullerene-based ETL device configuration [15]. Moreover, SAMs can particularly improve the interface between the active layer and the QD-HTL (PbS-EDT), rather than acting as an independent HTL [24]. This oxidation of PbS-EDT layer not only induces p-type conductivity by forming Pb–O and S–O species but also exposes undercoordinated Pb^{2+} sites through partial EDT desorption or rearrangement. These newly exposed or oxidized sites enable strong chemical bonding with SAM molecules like [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz) and [2-(3,6-dimethoxy-9H-carbazol-9-yl)ethyl]phosphonic acid (MeO-2PACz) via phosphonic acid groups, forming Pb–O–P linkages. Even in the presence of residual EDT, co-adsorption is possible, resulting in a mixed-ligand interface that supports stable and effective bonding of the SAMs on PbS-EDT layer [25]. In addition, C_{60} and [6,6]-phenyl-C61-butyric acid methyl ester (PCBM) are used together to create a cascade energy alignment, where C_{60} ensures efficient electron extraction and PCBM provides a smooth, compact transport layer to improve device stability and reduce leakage [26, 27]. To elucidate the underlying mechanisms, we tested the top contact angles of the HTLs, and measured QD stacking configuration of the QD active layer deposited on different HTLs by grazing-incidence small-angle X-ray scattering (GISAXS). Time-resolved photoluminescence (TR-PL) measurement revealed an enhanced electron blocking function by the SAM. Device performance demonstrated that 2PACz enhances charge carrier extraction, leading to an improved external quantum efficiency of over 53% at 1200 nm. Additionally, MeO-2PACz effectively suppresses the dark current density to 220 nA/cm^2 under a −0.5 V bias while increasing the specific detectivity to 1.64×10^{12} Jones of the inverted QD photodetectors.

2 Experimental section

2.1 Chemicals

All chemicals were used as received without further purification. Oleic acid (OA, 90%) was obtained from ThermoFisher. n-Octane (96.0%) and acetonitrile (ACN, 99.9%) were purchased from Macklin. N,N-Dimethylformamide (DMF, 98.0%) was obtained from J&K Scientific. Lead bromide ($PbBr_2$, 99.999%), lead iodide (PbI_2, 99.999%), nickel oxide (NiO_x, 99.999%), fullerene-C_{60} (C_{60}, 99.9%), PCBM (99.9%), bathocuproine (BCP, 99.5%), and chlorobenzene (99.9%) were sourced from Youxuan Technology. EDT (99.8%) and n-butylamine (99.0%) were acquired from Aladdin. 2PACz (98%) and MeO-2PACz (98%) were purchased from TCI. Ethanol (99.7%) was obtained from Rhawn.

2.2 Synthesis of QDs and QD ink fabrication

QD fabrication followed the procedure described in our previous work [28]. For QD ink fabrication, PbI_2 (1.84 g) and $PbBr_2$ (0.64 g) were dissolved in 15 mL of DMF. Oleic acid-capped PbS QDs were dissolved in n-octane (10 mg/mL) to prepare a 15 mL QD precursor solution, which was stirred vigorously with the DMF phase for 3–5 min until the QDs fully transferred to DMF. The mixture was allowed to settle for 1–3 min, and the supernatant was discarded. The QDs were washed three times with 10 mL of n-octane, followed by precipitation with 1 mL of n-octane and centrifugation at 7800 rpm for 3 min. The QD precipitate was dried for 40 min, yielding ligand-exchanged PbS-PbX_2 (X = I, Br) QDs. The precipitate was redispersed in a mixed DMF/n-butylamine solution at 350 mg/mL. The active layer was deposited via single-step spin-coating at 2500 rpm for 40 s, achieving a film thickness of ~ 350 nm. The film was annealed at 85 °C for 10 min in nitrogen and cooled to room temperature.

2.3 Photodetector fabrication

ITO glass substrates were sequentially ultrasonicated for 10 min each in a soap solution, de-ionized water, isopropanol, acetone, and ethanol. Immediately before NiO_x-HTL deposition, the substrates were treated with ultraviolet-ozone for 15 min. NiO_x solution was spin-coated onto ITO at 2000 rpm for 30 s in air, followed by annealing at 150 °C for 30 min and cooling to room temperature. PbS QD solution was spin-coated onto the NiO_x layer at 3000 rpm for 30 s, with EDT solution drop-cast for 30 s to perform solid-state ligand exchange. The film was spun at 4000 rpm for 10 s and rinsed with ACN at the same speed to remove oleic acid ligands. This process was repeated twice, followed by oxidation in air for at least 8 h to form the HTL. SAM solution was spin-coated onto the HTL at 4000 rpm for 10 s, annealed at 100 °C for 10 min, and cooled to room temperature. Ligand-exchanged PbS-PbX_2 (X = I, Br) QDs were spin-coated onto the SAM layer at 2500 rpm for 40 s, annealed at 85 °C for 10 min in nitrogen, and cooled to room temperature to form the active layer. A 10 nm C_{60} layer was thermally evaporated onto the active layer, followed by spin-coating PCBM (2000 rpm, 30 s) and BCP (4000 rpm, 30 s) solutions to form the electron transport layer. Finally, a 100 nm Ag top electrode was deposited by thermal evaporation in a vacuum chamber, completing the device.

2.4 Characterizations

The absorption spectrum of PbS QDs for the active layer was measured using an ultraviolet–visible-near-infrared spectrophotometer (Lambda 1050+, PerkinElmer). Contact angles of functional layer films were determined with a contact angle tester. Surface morphology was characterized by using a field-emission scanning electron microscope (Gemini SEM 300, Zeiss). Photoluminescence (PL) and TR-PL spectra were obtained using an X-ray fluorescence spectrometer (Spectro Midex, Spectro Analytical Instruments). GISAXS measurements were conducted on a SAXS machine XEUSS 2.0 with an X-ray photon energy of 8.05 keV (wavelength of 1.54 Å), a sample-to-detector distance (SDD) of 1003 mm, and an incident angle of 0.4°. Pilatus 3R 300K was used as the two-dimensional (2D) detector with a pixel size of 172 µm. EQE, responsivity, specific detectivity, and thin-film transistor (TFT) imaging performance of PbS QD photodetectors were evaluated using PD-QE and APD-QE systems (ENLITECH). Current–voltage (I–V) characteristics of photovoltaic detectors were measured using a Keithley 4200 digital source-meter. The active area of the photodetectors was determined to be 0.045 cm^2. The method for measuring the device's response time is as follows: a signal generator produces a 1 kHz square wave with a 50% duty cycle to drive a 1310 nm laser that illuminates the device's active region; the current signal is converted to a voltage signal by a current amplifier; with a bias of 0 V, the waveform is observed on an oscilloscope to obtain the response time. The −3 dB bandwidth was obtained with the same configuration; the current signal is converted to a voltage signal by a current amplifier; the waveform amplitude is read on an oscilloscope while the square-wave frequency is gradually increased from 1 Hz; when the optical response decreases to 0.707 times its initial amplitude, the corresponding frequency is taken as the −3 dB bandwidth.

3 Results and discussion

The morphology and absorption spectrum of the QDs used in the active layer are shown in Fig. S1 in the Electronic Supplementary Material (ESM), where the OA-capped QDs dispersed in octane exhibit an exciton peak around 1250 nm. OA-capped QDs with exciton peak at around 850 nm are used for the QD HTL fabrication. For the SAMs binding on the QD HTL, phosphoric acid groups (–PO_3H_2) can bind the Pb^{2+} ions due to Lewis acid–base coordination. Especially when the PbS surface has undergone partial oxidation forming Pb–OH or Pb–O^- groups, these oxygen-containing groups can undergo condensation reactions with phosphonic acid, resulting in the formation of stable P–O–Pb bonds [25, 29]. Morphologies of different HTLs are shown in Fig. S2 in the ESM. Figure 1(a) illustrates the contact angles of the HTLs, where the pristine PbS-EDT QD HTL exhibits 40.9°. After coating with SAMs, the contact angle increases to 65.5° for

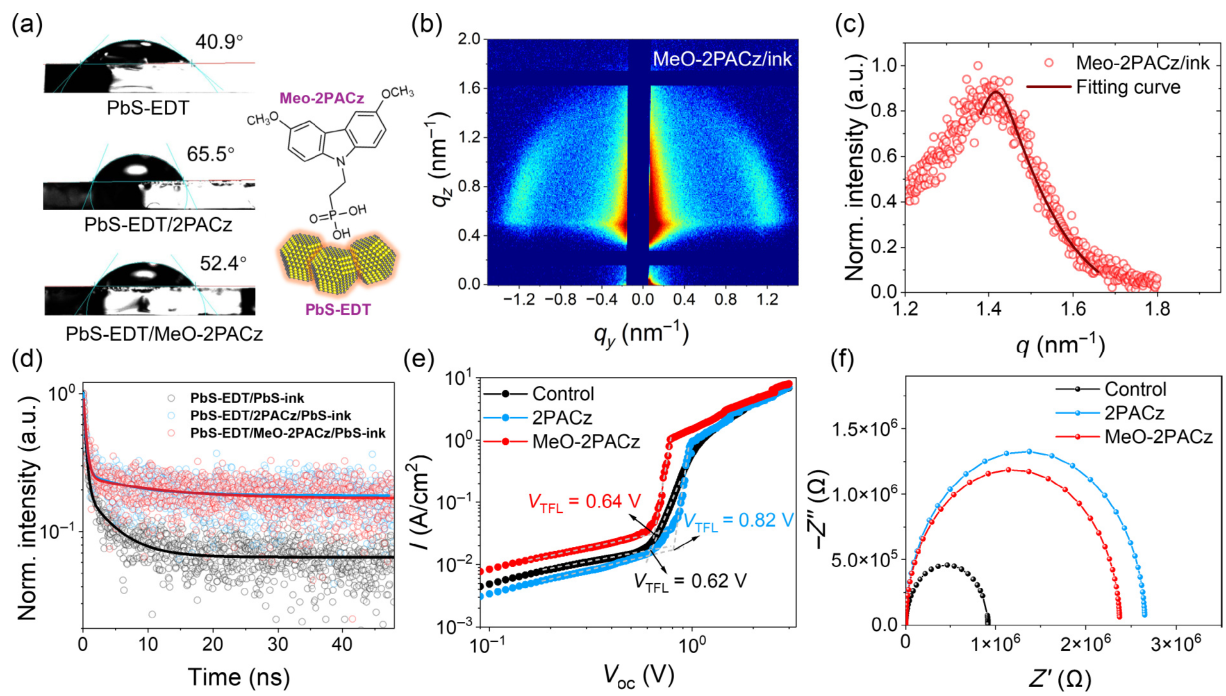

Figure 1 (a) Contact angles of different HTL surfaces. (b) GISAXS pattern of QD ink solid on MeO-2PACz. (c) Azimuthal integrations of the QD ink solid on MeO-2PACz. (d) TR-PL of the QD ink solids quenched by different HTLs. (e) V_{TFL} of the QD ink solids according to SCLC analysis. (f) Nyquist plots of the EIS data.

2PACz and 52.4° for MeO-2PACz respectively. The results are in good agreement with our expectations since the hydrophobic groups of 2PACz (such as methyl) dominate their high contact angle, while the methoxy group of MeO-2PACz reduces hydrophobicity through polar interactions. We confirmed that the morphologies of QD ink films deposited on different HTLs are very similar, as indicated in Fig. S3 in the ESM. To further investigate whether variations in contact angle influence the stacking kinetics of QDs in the active layer, thereby affecting the internal nanostructure, we conducted GISAXS measurements using an in-house small-angle X-ray scattering (SAXS) setup [28, 30]. Detailed measurement parameters are provided in the Experimental section. An incidence angle of 0.4° was selected, which exceeds the critical angle of the PbS film, ensuring sufficient X-ray penetration for comprehensive structural analysis of the entire QD film [31]. Figures 1(b) and 1(c) show the GISAXS pattern of the QD ink deposited on the HTL coated by MeO-2PACz, and the corresponding azimuthal integrations with fitting curves based on a hard-sphere model in a para-crystal arrangement [31]. The compared GISAXS patterns and fitting curves are shown in Fig. S4 in the ESM. All compared GISAXS patterns exhibit distinct characteristic scattering signals from strong-electronic coupled QDs [32, 33]. The fitting results, shown in Table S1 in the ESM, indicate that the QDs deposited on 2PACz (with the highest surface polarity) exhibit the smallest inter-dot distance of 4.47 ± 0.26 nm, suggesting a highest electronic coupling configuration.

Charge carrier dynamics to HTLs are initially analyzed via PL and TR-PL. PL spectra and the TR-PL fitting results are seen in Fig. S5 and Table S2 in the ESM. In the PL intensity comparison, both SAMs can quench the PbS-EDT layer due to the transfer of hole carriers to the SAMs, which reduces the PL intensity and carrier lifetime. In particular, MeO-2PACz, with its better conductivity, can more effectively quench the QD HTL compared to 2PACz [34]. We used a bi-exponential decay function to fit the PL decay dynamics, in which τ_1 indicates energy transfer to the interface and τ_2 stands for the trap state assisted radiative recombination. With 2PACz coated on PbS-EDT, τ_1 is increased, suggesting that the energy transfer between PbS-EDT and 2PACz is slightly hindered. This trend is reversed in the case of MeO-2PACz, which suggests a more efficient charge carrier transfer from the QD HTL to MeO-2PACz. We further analyze the charge carrier dynamics of the QD ink active layers on different HTL configurations by PL (as seen in Fig. S6 in the ESM) and TR-PL as seen in Fig. 1(d). The fitting results, shown in Table S3 in the ESM, suggest SAMs improve the PL lifetime of the active layer which we attribute to their ability to better block electron transfer from the active layer to the HTLs. Moreover, we further noticed that τ_2 values in all QD ink layers on SAM modified HTLs apparently increased, which suggests that SAMs can reduce the trap-state-induced radiative recombination of the QD ink layer at the contact interface.

Moreover, we fabricated single-carrier devices with the structures ITO/NiO_x/EDT/Ag and ITO/NiO_x/EDT/SAMs/Ag, respectively, to investigate charge carrier transport using space-charge-limited current (SCLC) analysis. The results, seen in Fig. 1(e), suggest SAMs increase V_{TFL} value to 0.82 V by 2PACz and 0.64 V by MeO-2PACz compared to pristine QD HTL (V_{TFL} = 0.62 V). Besides the trap states at the interface, we assume that SAMs introduce a higher energy barrier for hole transport in single carrier devices compared to pure QD HTL, in which 2PACz possesses a higher energy barrier than MeO-2PACz. We further analyze the charge carrier transfer in device level configuration via electrochemical impedance spectroscopy (EIS). The results of the EIS analysis of the device with different HTLs under a dark condition suggest that 2PACz exhibits the highest R_{ct} due to a higher energy barrier for electron transfer from active layer to ETL. The impedance response was fitting using an equivalent circuit as seen in Fig. S7 in the ESM and parameters were listed in Table S4 in the ESM [35].

The device architecture is given in Fig. 2(a) in which the NiO_x and PbS-EDT act as double HTLs, with the SAMs as the additional HTL. The device with pristine NiO_x and QD HTL is denoted as the control device and with additional SAM layer referred to by the name of the specific SAM in the following discussions. The J–V curves of the different HTL configured devices under dark and light conditions are shown in Fig. 2(b). Based on the current density in light condition, we confirmed that SAMs' introduction improves the charge extraction as indicated from the higher light current signal at zero bias. To quantitatively analyze the dark current, we used a dark current model, consisting of diode reverse saturated current (J_0), an ohmic component (J_{ohm}) , and a trap- or energy-barrier-induced non-ohmic component ($J_{non\text{-}ohm}$), to fit the dark J–V curves and distinguish the current contributions from different mechanisms, as described by Eq. (1) [36–38]

$$J_D = J_0\left(\mathrm{e}^{\frac{e}{nk_BT}(V-J_DR_S)}-1\right)+\frac{V-J_DR_S}{R_{SH}}+k(V-J_DR_S)^m \quad (1)$$

Here, J_D represents the dark current density of the device, V is the applied bias, and J_0 is the reverse saturation current density of the diode. R_S and R_{SH} denote the series resistance and shunt resistance of the device, respectively. N is the ideality factor of the diode, k represents the defect energy level, and m is the defect concentration. The fitting curves are demonstrated in Fig. 2(c) and Fig. S8 in the ESM. The fitting results are shown in Table S5 in the ESM. The results suggest that SAMs successfully decreased J_D at a bias of −0.5 V. In particular, the reduction in J_{ohm} indicates that SAMs improved the interface quality by decreasing unexpected leakage current. A reduced $J_{non\text{-}ohm}$ suggests reduced trap-induced or charge carrier accumulation-induced carrier recombination at the interface. Nonetheless, we noticed that the diode component increased slightly, which is attributed to the SAM-introduced energy barrier at the interface.

We further evaluate the device's performance in terms of multiple aspects, including response speed, −3 dB bandwidth, and noise spectral density. The corresponding results are presented in Figs. 2(d)–2(f). The response speeds were determined by measuring the time intervals for the current density to rise or fall between 10% and 90% of the peak intensity, both on the rising and falling edges, respectively, under excitation by a nanosecond-level laser. As listed in Table S6 in the ESM, the response speeds at the rise edge of the compared devices are similar, while the SAMs, particularly MeO-2PACz, slightly decrease the response time at the fall edge of the inverted device. Moreover, the control device exhibits the −3 dB bandwidth of 48 kHz. Both SAMs increase the values to 56 kHz for 2PACz and 62 kHz for MeO-2PACz. The improvement in −3 dB bandwidth is primarily attributed to enhanced carrier transport efficiency. On this basis, the transit time of carriers through the active layer is accelerated, leading to a reduced response time. Besides, interface passivation effectively reduces the density of defect states, suppressing non-radiative recombination of carriers. As a result, the device exhibits improved response speed and

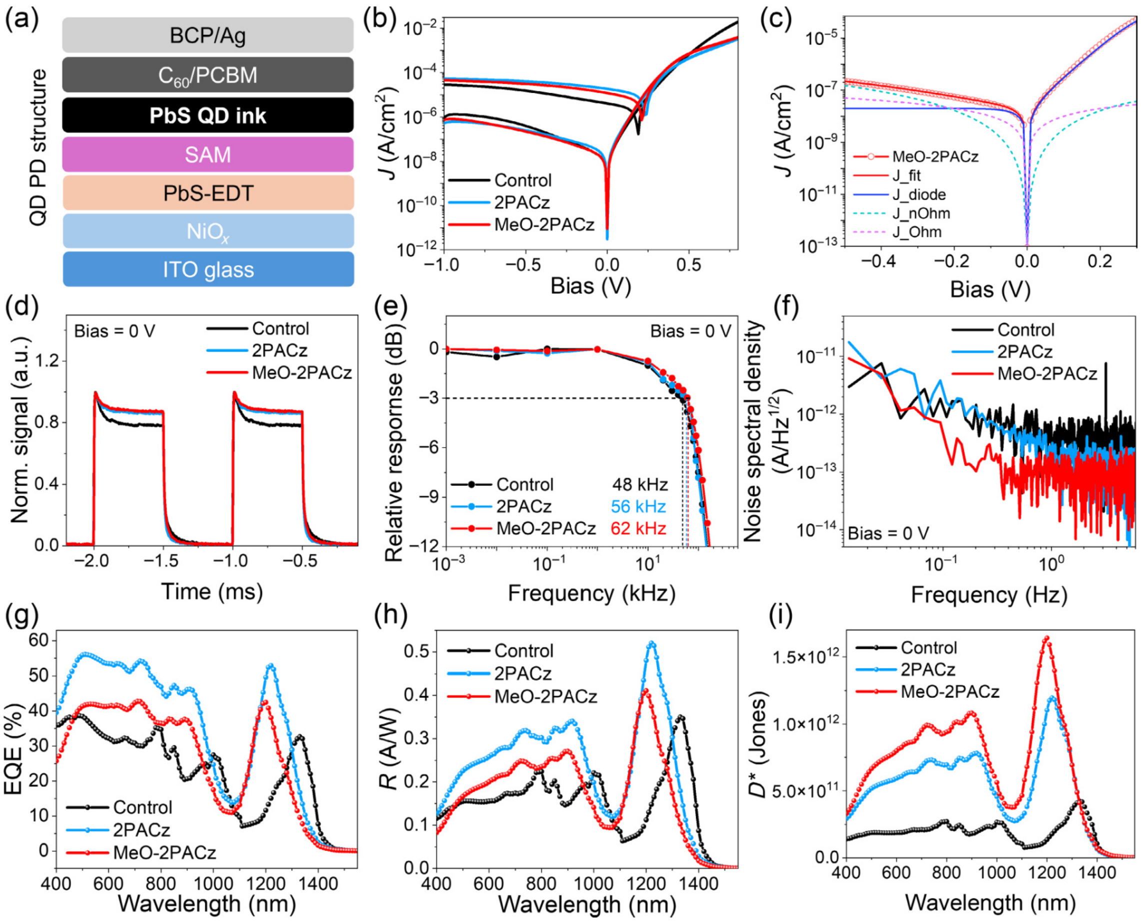

Figure 2 (a) Architecture of SAM induced devices. (b) J–V curves of devices with different SAM conditions. (c) Dark J–V fitting curves of the device with MeO-2PACz as additional HTL. (d) Response speed test results (illumination wavelength of 1310 nm). (e) 3 dB bandwidth and (f) noise spectral density test results of the compared devices. Spectra of (g) EQE, (h) R, and (i) D^* of the compared devices under 0 bias.

bandwidth performance. In addition, from the noise spectrum density, we confirm that SAMs, particularly MeO-2PACz, can significantly reduce the noise current density, which we attribute to a modified interface between the active layer and HTL. Figures 2(g)–2(i) show different spectra of responsivity (R), EQE and specific detectivity (D^*) of the compared devices with different HTL configurations. Based on the results, 2PACz can improve the charge carrier extraction and demonstrate the highest EQE of 53%. Notably, the EQE spectrum is determined by a calibrated instrument with standard detectors. R is thus obtained from Eq. (2)

$$R = \frac{\lambda}{hc}\mathrm{EQE} \tag{2}$$

in which λ stands for the wavelength, h is plank constant, and c is light speed. The responsivity spectra of the devices are shown in Fig. 2(g). On this basis, D^* is determined by Eq. (3) as

$$D^* = \frac{(A\,\Delta f)^{\frac{1}{2}}R}{i_n} \tag{3}$$

where A is the area for dark current density (0.045 cm^2) and Δf is set to 1 Hz. i_n is the dark current read from the noise spectral density at 1 Hz [32, 39]. Notably, the MeO-2PACz-based photodetector exhibits the highest D^* of 1.64×10^{12}, which is almost 4 times greater than the value of the control device. Table S7 in the ESM summarizes the primary parameters of the device performance. Moreover, we confirm that this value actually surpasses most state-of-the-art inverted PbS QD devices operating beyond 1100 nm under comparable testing conditions, as reported in Refs. [1, 2, 8, 10]. A detailed comparison is presented in Table S8 in the ESM. Although our device structure incorporates multiple HTLs and fullerene-based ETLs, resulting in additional interfaces, it effectively suppresses dark current and enhances EQE. We observe a blue shift of the excitonic peak in the EQE spectra of SAM-modified devices. We attribute this behavior to the interfacial dipole introduced by the SAM layer: its molecular dipoles are oriented from the QD layer toward the bottom electrode, creating an electric field that opposes the device's built-in field [25]. This counter-field lifts the QD energy levels with respect to the vacuum level, increasing the extraction barrier for carriers generated by low-energy (long-wavelength) photons. Consequently, the long-wavelength EQE is suppressed, while the response to higher-energy photons is relatively enhanced, shifting the EQE onset toward shorter wavelengths. Notably, the improved surface passivation provided by MeO-2PACz contributes to the highest D^* achieved to date.

To determine the range of light intensities over which a photodetector or imaging device can respond linearly and accurately without distortion, saturation, or noise interference, we have carried out J–V measurement under different light conditions. The results are shown in Fig. 3(a) and Fig. S9 in the ESM. The derived linear dynamic range (LDR) is further given in Fig. 3(b).

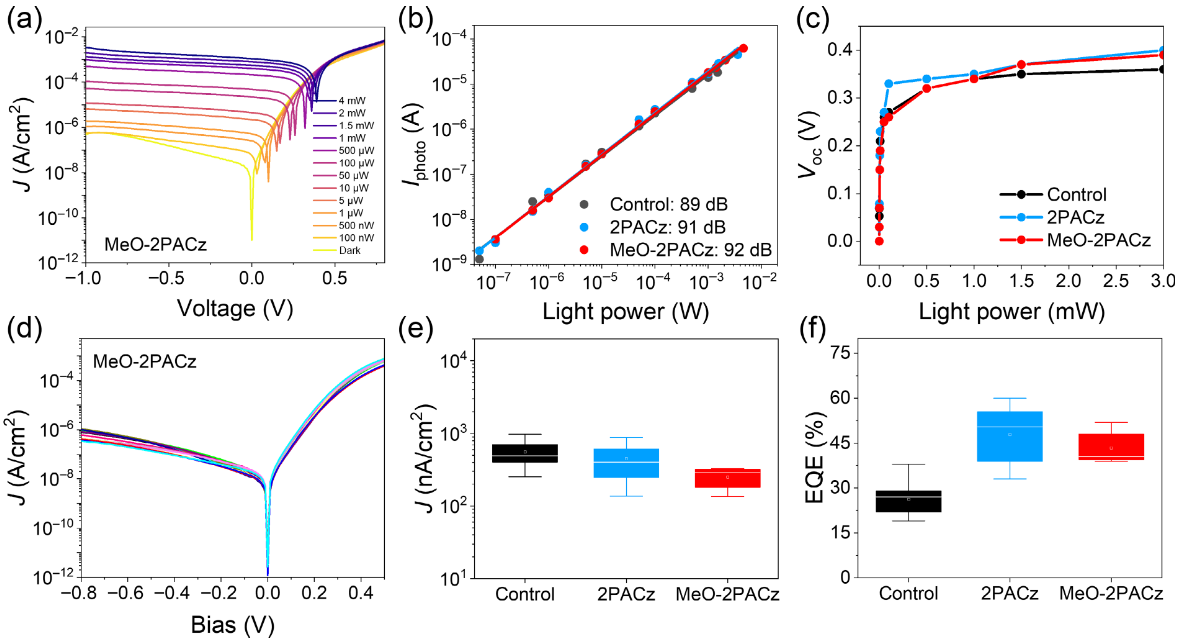

Figure 3 (a) J–V curves of device under different power illumination with wavelength of 1310 nm. (b) LDR fitting results of the compared devices in which I_{photo} denotes the photocurrent. (c) V_{oc} of the devices along with different light power. (d) Uniformity test of J–V characters of the devices in dark condition. (e) Statistics of the dark current densities of the compared devices. (f) Statistics of the EQE of the compared devices.

According to the linear fitting, SAMs can improve the LDR of the device with MeO-2PACz achieving the highest LDR of 92 dB. We further tracked the variation of the open circuit voltage (V_{oc}) of the devices under the modulated light conditions. We confirm that 2PACz endows the highest V_{oc} at a small light condition, which suggests that the 2PACz can better passivate the interface and reduce the recombination. SAMs enable higher V_{oc}'s of the devices eventually as the illumination intensity increases. In addition, to verify the repeatability of SAMs improved device performance particularly with respect to J_D (bias = −0.5 V) and EQE (bias = 0 V), we measured these parameters of over 54 devices and presented derived data in Figs. 3(d) and 3(e). MeO-2PACz devices exhibit narrowly distributed J_D values as indicated in Fig. 3(d). The J_D and EQE distributions remain consistent with the afore-mentioned performance, confirming the excellent repeatability and reliability of the SAM-modified devices.

To demonstrate the compatibility of our device configuration with most read-out circuits and to demonstrate its suitability for SWIR imaging, we fabricated device arrays on an existing TFT substrate (12 pixels × 12 pixels), as shown in Fig. 4(a). The device architecture is identical to the previously described devices, with NiO_x/PbS-EDT/MeO-2PACz serving as the HTLs and C60/PCBM as the ETLs. To selectively mask the pixel array under a 1310 nm light source, we used different letter-shaped masks, as shown in Fig. 4(b). As a result, we successfully generated and composed clearly distinguishable "SZTU" patterns, as indicated in Fig. 4(c). Notably, this device architecture is also compatible with CMOS-based ROIC which has the potential to provide higher resolution and better imaging quality.

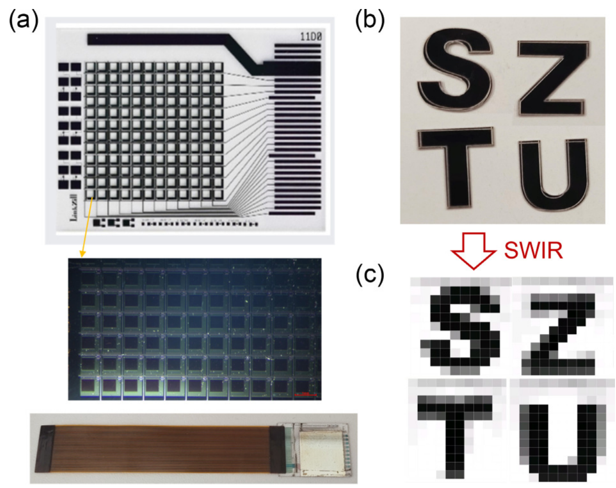

Figure 4 (a) A 12 pixels × 12 pixels TFT substrate with existing read-out circuits was employed. (b) Fabricated letter-shaped masks. (c) Images obtained from the device array under SWIR illumination.

4 Conclusion

We have explored an inverted photodetector device compatible with readout circuits and based on multi-HTLs by introducing different SAMs as additional HTLs on NiO_x and PbS-EDT. Photophysics studies suggest SAMs can passivate oxidized surface of thiol-treated PbS QDs. The SAM-modified photodetectors exhibit significantly improved performance, including an EQE exceeding 40% and reduced dark current density. Specifically, 2PACz more effectively reduces surface recombination, thereby enhancing charge extraction, while MeO-2PACz reduces the leakage current by minimizing ohmic leakage, thus achieving a specific detectivity of 1.6 × 10^{12} Jones, comparable to that of commercial photodetectors for SWIR imaging.

Electronic Supplementary Material: Supplementary material (TEM and absorption spectra of QDs, SEM and GISAXS analyses, spectral measurements and TRPL of comparative QD solids with corresponding fitting results, EIS equivalent circuit and fitting results, device performance including J–V curves with fitting results, response speed, and a comparison table of our device performance with that of devices reported in recent literature) is available in the online version of this article at https://doi.org/10.26599/NR.2025.94907796.

Data availability

All data needed to support the conclusions in the paper are presented in the manuscript and the Electronic Supplementary Material. Additional data related to this paper may be requested from the corresponding author upon request.

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Declaration of competing interest

All the contributing authors report no conflict of interests in this work.

Author contribution statement

S. M. C.: Investigation, data curation, writing – original draft. H. B. Z.: Investigation, data curation. S. C.: Investigation, data curation. H. D. T.: Investigation, methodology, writing – review & editing. X. J. Z.: Investigation. Q. C.: Investigation. Y. H. S.: Investigation. Y. M. C.: Investigation. Y. L.: Investigation. Y. H. T.: Investigation. F. F.: Investigation. J. J. H.: Investigation. Z. G. T.: Resources, investigation. K. Y. Z.: Resources, investigation. W. C.: Conceptualization, supervision, writing – review & editing, funding acquisition.

Use of AI statement

None.

References

[1] Wang, Y.; Liu, J.; Ran, X.; Lu, H. L.; Liu, J.; Yang, J. R.; Xia, H.; Zhang, X. C.; Zhang, J. B.; Gao, L. et al. Inorganic ZnTe hole transport layer for PbS quantum dot infrared photodetectors and imagers. *Adv. Funct. Mater*., in press, DOI: 10.1002/adfm.202426082.

[2] Manders, J. R.; Lai, T. H.; An, Y. B.; Xu, W. K.; Lee, J.; Kim, D. Y.; Bosman, G.; So, F. Low-noise multispectral photodetectors made from all solution-processed inorganic semiconductors. *Adv. Funct. Mater.* **2014**, *24*, 7205–7210.

[3] Wang, Y. J.; Wu, H.; Rodà, C.; Peng, L. C.; Taghipour, N.; Dosil, M.; Konstantatos, G. Shortwave infrared light detection and ranging using silver telluride quantum dots. *Adv. Mater.* **2025**, *37*, 2500977.

[4] Yu, M. X.; Huang, R. Z.; Zhang, X. C.; Hu, X. D.; Ran, X.; Xia, H.; Yuan, M. H.; Zhang, J. B.; Gao, L.; Tang, J. et al. Gold doping of HgTe colloidal quantum dots for high-performance infrared photodetectors. *Adv. Funct. Mater*., in press, DOI: 10.1002/adfm.202423299.

[5] Mu, G.; Zheng, X. L.; Tan, Y. M.; Liu, Y. F.; Hao, Q.; Weng, K. K.; Tang, X. Colloidal quantum-dot heterojunction imagers for room-temperature thermal imaging. *Adv. Mater.* **2025**, *37*, 2416877.

[6] Mu, G.; Tan, Y. M.; Bi, C.; Liu, Y. F.; Hao, Q.; Tang, X. Visible to mid-wave infrared PbS/HgTe colloidal quantum dot imagers. *Nat. Photon.* **2024**, *18*, 1147–1154.

[7] Zhang, S.; Bi, C.; Qin, T. L.; Liu, Y. F.; Cao, J.; Song, J. Q.; Huo, Y. J.; Chen, M. L.; Hao, Q.; Tang, X. Wafer-scale fabrication of CMOS-compatible trapping-mode infrared imagers with colloidal quantum dots. *ACS Photonics* **2023**, *10*, 673–682.

[8] Zhang, L. X.; Chen, L.; Yang, J. R.; Liu, J.; Lu, S. C.; Liang, X. Y.; Zhao, X. Z.; Yang, Y.; Hu, J.; Hu, L. et al. High-performance and stable colloidal quantum dots imager via energy band engineering. *Nano Lett.* **2023**, *23*, 6489–6496.

[9] Liu, J.; Liu, P. L.; Chen, D. Y.; Shi, T. L.; Qu, X. X.; Chen, L.; Wu, T.; Ke, J. P.; Xiong, K.; Li, M. Y. et al. A near-infrared colloidal quantum dot imager with monolithically integrated readout circuitry. *Nat. Electron.* **2022**, *5*, 443–451.

[10] Parmar, D. H.; Rehl, B.; Atan, O.; Hoogland, S.; Sargent, E. H. Transient measurements and simulations correlate exchange ligand concentration and trap states in colloidal quantum dot photodetectors. *ACS Appl. Mater. Interfaces* **2023**, *15*, 59931–59938.

[11] Wang, B. B.; Yuan, M. H.; Liu, J.; Zhang, X. C.; Liu, J.; Yang, J.; Gao, L.; Zhang, J. B.; Tang, J.; Lan, X. Z. Synergism in binary nanocrystals enables top-illuminated HgTe colloidal quantum dot short-wave infrared imager. *Nano Lett.* **2024**, *24*, 9583–9590.

[12] Yin, S. C.; Ho, C. H. Y.; Ding, S.; Fu, X. Y.; Zhu, L. P.; Gullett, J.; Dong, C.; So, F. Enhanced surface passivation of lead sulfide quantum dots for short-wavelength photodetectors. *Chem. Mater.* **2022**, *34*, 5433–5442.

[13] Chen, L.; Liu, S. J.; Liu, Z. Y.; Wu, W. D.; Li, H.; Liu, Y. X.; Zhou, X.; Liu, J.; Lan, X. Z.; Li, L. Y. et al. Complementary passivation of bidentate aromatic ligands enables high-temperature stable PbS colloidal quantum dots image sensor. *Adv. Funct. Mater*., in press, DOI: 10.1002/adfm.202501770.

[14] Song, Z. L.; Cao, T.; Wang, X.; Tan, Y. Z.; Wu, J. F.; Wang, Z. J.; Zhu, H. B.; Fang, F.; Wu, D.; Chen, W. et al. Reducing dark current density of PbS quantum dot short-wave infrared photodetectors by polymer buffer layer modification. *IEEE Electron Device Lett.* **2024**, *45*, 2455–2458.

[15] Wang, R. L.; Wu, X.; Xu, K. M.; Zhou, W. J.; Shang, Y. Q.; Tang, H. Y.; Chen, H.; Ning, Z. J. Highly efficient inverted structural quantum dot solar cells. *Adv. Mater.* **2018**, *30*, 1704882.

[16] Wang, H.; Pinna, J.; Romero, D. G.; Di Mario, L.; Koushki, R. M.; Kot, M.; Portale, G.; Loi, M. A. PbS quantum dots ink with months-long shelf-lifetime enabling scalable and efficient short-wavelength infrared photodetectors. *Adv. Mater.* **2024**, *36*, 2311526.

[17] Wang, Y.; Hu, H. C.; Yuan, M. H.; Xia, H.; Zhang, X. C.; Liu, J.; Yang, J.; Xu, S. Q.; Shi, Z. R.; He, J. G. et al. Colloidal PbS quantum dot photodiode imager with suppressed dark current. *ACS Appl. Mater. Interfaces* **2023**, *15*, 58573–58582.

[18] Kirmani, A. R.; Sheikh, A. D.; Niazi, M. R.; Haque, M. A.; Liu, M. X.; de Arquer, F. P. G.; Xu, J. X.; Sun, B.; Voznyy, O.; Gasparini, N. et al. Overcoming the ambient manufacturability-scalability-performance bottleneck in colloidal quantum dot photovoltaics. *Adv. Mater.* **2018**, *30*, 1801661.

[19] Tang, H. C.; Shen, Z. C.; Shen, Y. Z.; Yan, G.; Wang, Y. B.; Han, Q. F.; Han, L. Y. Reinforcing self-assembly of hole transport molecules for stable inverted perovskite solar cells. *Science* **2024**, *383*, 1236–1240.

[20] Yang, X. K.; Ma, T. J.; Hu, H. J.; Ye, W. J.; Li, X.; Li, M. Y.; Zhang, A. F.; Ge, C. Y.; Sun, X. L.; Zhu, Y. X. et al. Understanding and manipulating the crystallization of Sn–Pb perovskites for efficient all-perovskite tandem solar cells. *Nat. Photon.* **2025**, *19*, 426–433.

[21] Jiang, Q.; Zhu, K. Rapid advances enabling high-performance inverted perovskite solar cells. *Nat. Rev. Mater.* **2024**, *9*, 399–419.

[22] Park, S. M.; Wei, M. Y.; Lempesis, N.; Yu, W. J.; Hossain, T.; Agosta, L.; Carnevali, V.; Atapattu, H. R.; Serles, P.; Eickemeyer, F. T. et al. Low-loss contacts on textured substrates for inverted perovskite solar cells. *Nature* **2023**, *624*, 289–294.

[23] Jiang, Q.; Tirawat, R.; Kerner, R. A.; Gaulding, E. A.; Xian, Y. M.; Wang, X. M.; Newkirk, J. M.; Yan, Y. F.; Berry, J. J.; Zhu, K. Towards linking lab and field lifetimes of perovskite solar cells. *Nature* **2023**, *623*, 313–318.

[24] Dantas Campos, R.; Singh, S.; Heffner, H.; Löffler, M.; Paulus, F.; Vaynzof, Y. Self-assembled molecules for hole extraction in efficient inverted PbS quantum dot solar cells. *J. Mater. Chem. A* **2024**, *12*, 32689–32696.
[25] Yu, M. X.; Wang, B. B.; Huang, R. Z.; Xia, H.; Liu, J.; Deng, C. J.; Zhang, J. B.; Gao, L.; Tang, J.; Lan, X. Z. Interfacially-engineered HgTe colloidal quantum dot photodiodes using self-assembled monolayer. *ACS Photonics* **2024**, *11*, 5100–5109.
[26] Lee, D.; Jung, Y. S.; Heo, Y. J.; Lee, S.; Hwang, K.; Jeon, Y. J.; Kim, J. E.; Park, J.; Jung, G. Y.; Kim, D. Y. Slot-die coated perovskite films using mixed lead precursors for highly reproducible and large-area solar cells. *ACS Appl. Mater. Interfaces* **2018**, *10*, 16133–16139.
[27] Kapil, G.; Bessho, T.; Sanehira, Y.; Sahamir, S. R.; Chen, M. M.; Baranwal, A. K.; Liu, D.; Sono, Y.; Hirotani, D.; Nomura, D. et al. Tin-lead perovskite solar cells fabricated on hole selective monolayers. *ACS Energy Lett.* **2022**, *7*, 966–974.
[28] Chen, S. M.; Zhong, H. Y.; Wang, X.; Pan, G. J.; Tang, H. D.; Fang, F.; Wu, J. F.; Wang, W. C.; Xu, L. H.; Tang, J. et al. Hybrid-size quantum dots in hole transport layer depress dark current density of short-wave infrared photodetectors. *ACS Photonics* **2025**, *12*, 879–888.
[29] Al-Ashouri, A.; Magomedov, A.; Roß, M.; Jošt, M.; Talaikis, M.; Chistiakova, G.; Bertram, T.; Márquez, J. A.; Köhnen, E.; Kasparavičius, E. et al. Conformal monolayer contacts with lossless interfaces for perovskite single junction and monolithic tandem solar cells. *Energy Environ. Sci.* **2019**, *12*, 3356–3369.
[30] Chen, W.; Zhong, J. L.; Li, J. Z.; Saxena, N.; Kreuzer, L. P.; Liu, H. C.; Song, L.; Su, B.; Yang, D.; Wang, K. et al. Structure and charge carrier dynamics in colloidal PbS quantum dot solids. *J. Phys. Chem. Lett.* **2019**, *10*, 2058–2065.
[31] Chen, W.; Guo, R. J.; Tang, H. D.; Wienhold, K. S.; Li, N.; Jiang, Z. Y.; Tang, J.; Jiang, X. Y.; Kreuzer, L. P.; Liu, H. C. et al. *Operando* structure degradation study of PbS quantum dot solar cells. *Energy Environ. Sci.* **2021**, *14*, 3420–3429.
[32] Liu, P. L.; Lu, S. C.; Liu, J.; Xia, B.; Yang, G. Y.; Ke, M.; Zhao, X. Z.; Yang, J. R.; Liu, Y. X.; Ge, C. Y. et al. Double-ended passivator enables dark-current-suppressed colloidal quantum dot photodiodes for CMOS-integrated infrared imagers. *InfoMat* **2024**, *6*, e12497.
[33] Sun, B.; Johnston, A.; Xu, C.; Wei, M. Y.; Huang, Z. R.; Jiang, Z.; Zhou, H.; Gao, Y. J.; Dong, Y. T.; Ouellette, O. et al. Monolayer perovskite bridges enable strong quantum dot coupling for efficient solar cells. *Joule* **2020**, *4*, 1542–1556.
[34] Lee, E. M. Y.; Tisdale, W. A. Determination of exciton diffusion length by transient photoluminescence quenching and its application to quantum dot films. *J. Phys. Chem. C* **2015**, *119*, 9005–9015.
[35] Cao, T.; Chen, S. M.; Fang, F.; Tang, H. D.; Hao, J. J.; Tang, J.; Cheng, J. J.; Chen, W. Size effects of 1,2-ethanedithiol-treated PbS quantum dots on short-wave infrared photodetector hole transport layers. *J. Phys. Chem. Lett.* **2025**, *16*, 4607–4614.
[36] Yan, Y.; Liu, H. Y.; Bian, L.; Dai, Y. Y.; Zhang, B. N.; Xue, S. M.; Zhou, Y.; Xu, J. L.; Wang, S. D. Origin and suppression of dark current for high-performance colloidal quantum dot short-wave infrared photodetectors. *Mater. Horiz.* **2024**, *11*, 6192–6221.
[37] Williams, B. L.; Smit, S.; Kniknie, B. J.; Bakker, K. J.; Keuning, W.; Kessels, W. M. M.; Schropp, R. E. I.; Creatore, M. Identifying parasitic current pathways in CIGS solar cells by modelling dark *J*–*V* response. *Prog. Photovoltaics: Res. Appl.* **2015**, *23*, 1516–1525.
[38] Wang, Y. J.; Peng, L. C.; Schreier, J.; Bi, Y.; Black, A.; Malla, A.; Goossens, S.; Konstantatos, G. Silver telluride colloidal quantum dot infrared photodetectors and image sensors. *Nat. Photon.* **2024**, *18*, 236–242.
[39] Fang, Y. J.; Armin, A.; Meredith, P.; Huang, J. S. Accurate characterization of next-generation thin-film photodetectors. *Nat. Photon.* **2019**, *13*, 1–4.